# ALEXANDER R. PELUSO

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## SUMMARY

Computer Science graduate (University of Delaware) who builds full-stack web apps with React, TypeScript, and FastAPI, with hands-on experience in REST APIs, data pipelines, automated testing, and AI-agent tooling. Seeking an entry-level role in web application development or software engineering. Eligible for U.S. Security Clearance.

## EDUCATION

**University of Delaware**
Bachelor of Science in Computer Science, Systems and Networks Concentration
Relevant Coursework: Software Engineering, Databases, Computer Networks, Operating Systems, Cybersecurity, Machine Learning

## PROJECTS

**LoL Survival Tracker (React, TypeScript, Tailwind, FastAPI, SQLite, Riot API) | In Progress** *2026*

- Built a full-stack web app to run a League of Legends challenge with friends (spin wheels, shared champion pool, eliminations, revive tokens, and stats) using React, TanStack Query, and a layout that works on mobile.
- Wrote a FastAPI backend with a background poller that pulls finished matches from the Riot API, checks them against the challenge rules, and applies results automatically, flagging edge cases for manual review.
- Set up GitHub Actions CI that runs 120+ backend and frontend tests (pytest, Vitest), a production build, and Alembic migration checks. Repo: github.com/apeluso03/LeagueSurvival

**Fantasy Football AI Agent & Desktop GUI (Python, OpenAI Responses API, SQLite, Tkinter) | In Progress** *2026*

- Built an AI agent for a 12-team superflex keeper league that answers keeper, trade, and draft questions by reasoning over live Sleeper data and scraped rankings through a hand-rolled tool-calling loop.
- Designed deterministic keeper-cost, trade-value, and power-ranking engines kept separate from the LLM, covered by 84 offline unit tests.
- Shipped two Windows desktop editions, including a user edition with a setup wizard that connects to any Sleeper league, plus player search and recent-performance snapshots. Repo: github.com/apeluso03/FantasyAgent

**FileFlow, Async CSV Processing REST API (FastAPI, Celery, Redis, PostgreSQL, Docker)** *2026*

- Built a 5-service containerized pipeline with async task queuing, API key auth, and a CI-tested integration suite (11 tests, no mocks). Repo: github.com/apeluso03/fileflow

**Custom Language Compiler (TypeScript) & Custom UNIX Shell (C)** *2023 - 2024*

- Built a compiler targeting WebAssembly and RISC-V and a UNIX shell with process creation, piping, and I/O redirection, with memory safety verified via Valgrind.

## WORK HISTORY

**Data Pipeline Developer, EPRI (Client Project)** *2024 - 2025*

- Built an ETL pipeline with Scrapy and Playwright to scrape course catalogs from 50+ universities, plus a keyword-scoring system that flagged hydrogen energy curriculum gaps in structured reports the client used for curriculum decisions.

**Shift Leader, Playa Bowls | Toms River, NJ / Newark, DE** *May 2024 - Present*

## ACTIVITIES

**Production Team Member, UD Esports** *Sep 2023 - May 2025*

- Operated live Twitch broadcasts (OBS, Stream Deck, GOXLR) and designed overlays/camera systems; awarded Producer of the Year (2024).

## SKILLS

**Programming Languages:** TypeScript, JavaScript, Python, SQL, HTML/CSS, C, C++, C#, Java
**Web & Frameworks:** React, Tailwind CSS, Vite, TanStack Query, FastAPI, REST APIs, SQLModel, Celery
**Tools & Technologies:** Git, GitHub Actions, Docker, PostgreSQL, SQLite, Redis, Alembic, pytest, Vitest, Playwright, Scrapy, Claude/OpenAI/Gemini APIs
**Core Competencies:** Full-Stack Web Development, API Design, Automated Testing, Data Pipelines, AI Agent Tooling, Debugging, Agile Development
**General:** Problem Solving, Communication, Collaboration, Adaptability